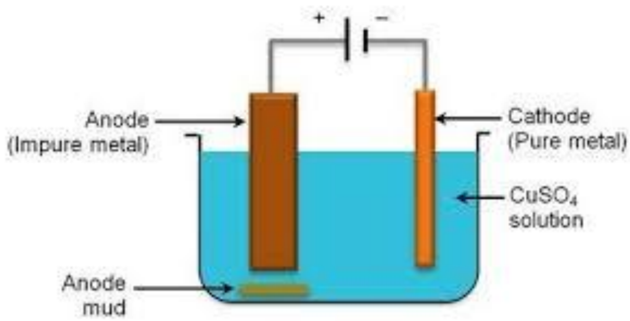
SEE 2082 (2026)
अनिवार्य विज्ञान तथा प्रविधि
उत्तरकुन्जिका

यस कुन्जिकामा भएको बाहेक अन्य सही वैकल्पिक तरिकाबाट समस्या समाधान गरेमा पनि अंक प्रदान गर्नुपर्ने छ । उत्तरकुञ्जिकामा प्रत्येक चरणको प्राप्तांङ्क १ प्रदान गर्नुपर्नेछ तथा एकाइ छुट भएमा ०.५ अङ्क मात्र घटाउनुपर्ने छ ।

<table>
<tr><th>प्र. नं.</th><th>उत्तर</th><th>अंक</th></tr>
<tr><td colspan="3">खण्ड ‘क’ (Section ‘A’)</td></tr>
<tr><td>1 a.</td><td>(ii) Amount of fertilizer</td><td>1</td></tr>
<tr><td>b</td><td>(iii) sending and receiving signals are done in the same channel simultaneously</td><td>1</td></tr>
<tr><td>c</td><td>(ii) by reducing the amount of carbon dioxide through photosynthesis</td><td>1</td></tr>
<tr><td>d</td><td>(i) 396 N</td><td>1</td></tr>
<tr><td>e</td><td>(iv) 500N</td><td>1</td></tr>
<tr><td>f</td><td>(i) thermal expansion</td><td>1</td></tr>
<tr><td>g</td><td>(iii) it increases AC voltage</td><td>1</td></tr>
<tr><td>h</td><td>(iv) $F < O < N$</td><td>1</td></tr>
<tr><td>i</td><td>(iv) Concentration, Oxidation, Reduction, Refining</td><td>1</td></tr>
<tr><td>j</td><td>(ii) Carbendazium</td><td>1</td></tr>
<tr><td colspan="3">खण्ड ‘ख’ (Section ‘B’)</td></tr>
<tr><td>2.a</td><td><table><tr><th>rare species</th><th>endangered species</th></tr><tr><td>a) their number are constant in the world</td><td>a) their number are decreasing drastically.</td></tr></table></td><td>1</td></tr>
<tr><td>b</td><td>- It helps in developing immunity
- it provides Physical energy
- it helps to increase sexual stamina.</td><td>1</td></tr>
<tr><td>c</td><td>- The substance that wants to float on liquid should displace the liquid equal to its weight. (the weight of floating object is equal to the weight of liquid displaced by it.)</td><td>1</td></tr>
<tr><td>d</td><td>- Anomalous behavior of water</td><td>1</td></tr>
<tr><td>e</td><td>- They are inversely proportional to each other</td><td>1</td></tr>
<tr><td>f</td><td>- they have same number of valence electron / same number of electrons in their outermost shell</td><td>1</td></tr>
<tr><td>g</td><td>- $CaCO_3$ ------ $CaO + CO_2$</td><td>1</td></tr>
<tr><td>h</td><td>

-</td><td>1</td></tr>
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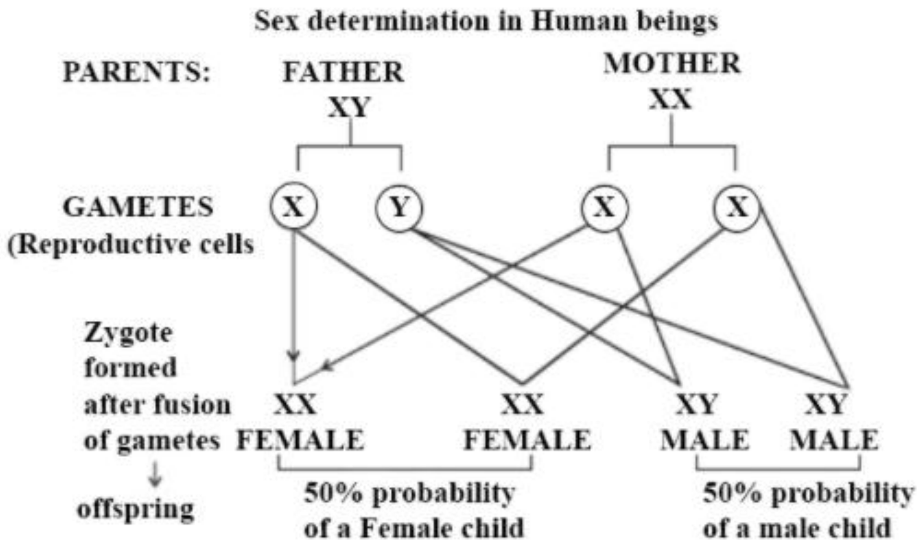
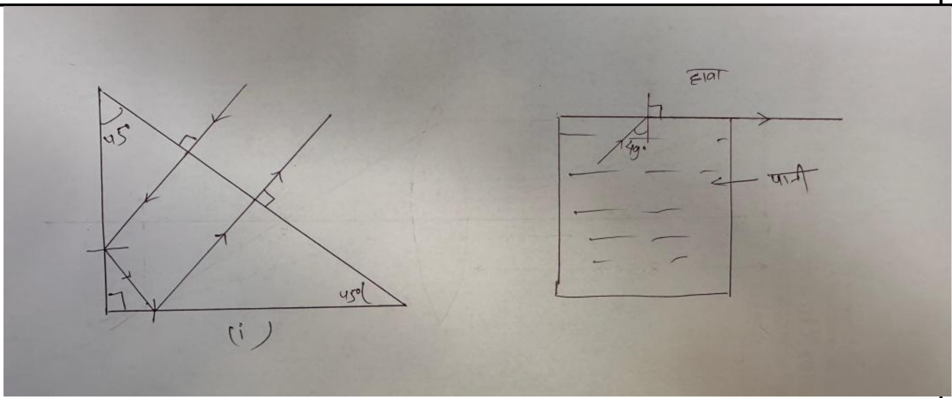
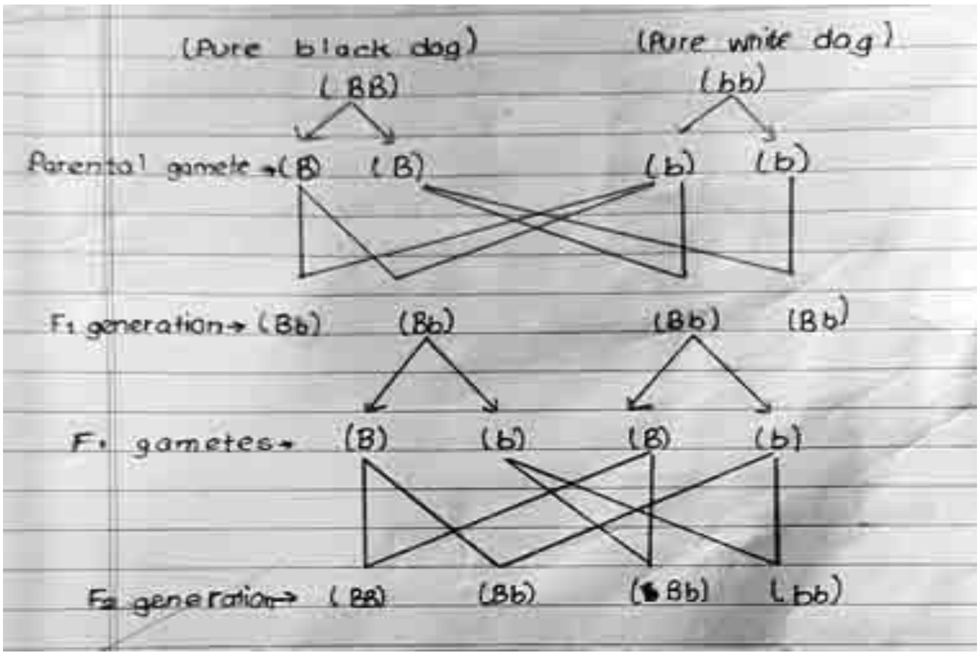
<table>
<tr><td>2 i</td><td>- butane</td><td>1</td></tr>
<tr><td colspan="3">खण्ड ‘ग’ (Section ‘C’)</td></tr>
<tr><td>3</td><td>- $v = u + at^2$
LHS = V
= m/s
RHS= $u + at^2$
= $m/s + m/s^2 \times s^2$
= m/s + m

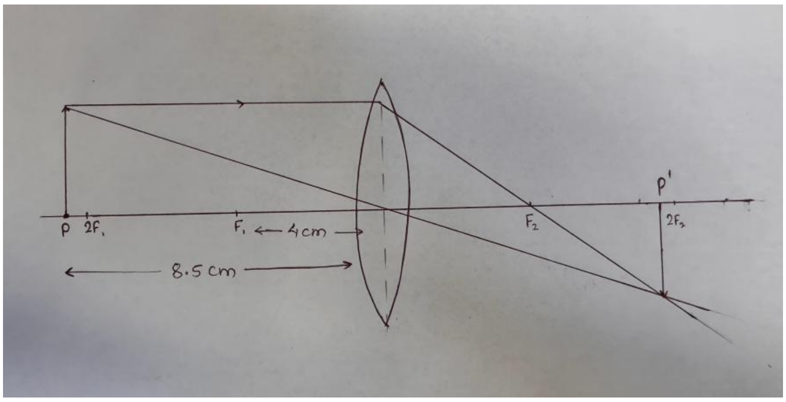
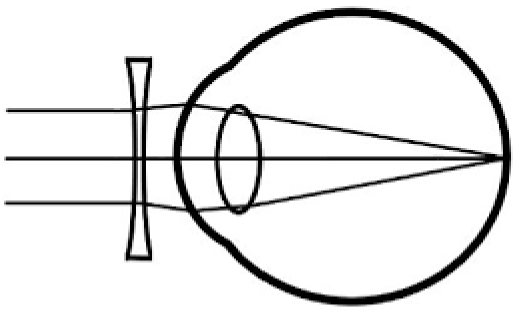
conclusion: the left hand side is not equal to right hand side, so the equation is invalid.</td><td>1

1</td></tr>
<tr><td>4</td><td><table><tr><td>horse (Fig A)</td><td>Sea Horse (fig (B)</td></tr><tr><td>- it respires through lungs
- it is viviparous</td><td>- it respires through gills
- it is oviparous</td></tr></table></td><td>1
1</td></tr>
<tr><td>5</td><td>- it has vascular tissue (Xylem and phloem)
- the plant body is divided into root, stem and leaf</td><td>1 for each</td></tr>
<tr><td>6</td><td>P- drone
Q- queen / worker</td><td>1 mark for each</td></tr>
<tr><td>7</td><td><table><tr><td>mitosis</td><td>meiosis</td></tr><tr><td>1. one mother cell divides into two daughter cell
2. number of chromosomes remains same both in mother and daughter cells
3. it occurs in somatic/ vegetative cells
4. it helps in growth and development</td><td>1. one mother cell divides into four daughter cell
2. number of chromosome becomes half in daughter cells
3. it occurs in reproductive cells
4. it helps in sexual reproduction</td></tr></table></td><td>2 marks for any two points.</td></tr>
<tr><td>8</td><td>
</td><td>1

1</td></tr>
<tr><td>9</td><td>a) A negative
b) Antigen B</td><td>1
1</td></tr>
</table>

10	Given; m = 500gm = 0.5 kg $t_1 = 30℃$ $t_2 = 90℃$ s = 4200 J/kg℃ Q= ? we have, Q= msdt Q = 0.5 × 4200 ×60 Q= 126000J The amount of heat energy required by the water is 126000J	1 1
11	 	1 1
12	- The heavenly bodies are moving away from the center - the size of universe is expanding	1 1
13	a) VI A / 16 b) it loses electron during chemical reaction / it is electropositive in nature	1 1
14	- zinc gets eroded from the bottle wall by the reaction with hydrochloric acid. As a result, the wall gets thinner. $Zn + 2HCl$ -------- $ZnCl_2 + H_2$	1 1
15	a. C_2H_5OH b. It lies in alcohol group.	1 1
16	- due to open vessel, the pickle got contact with oxygen and moisture and development of fungi and bacteria occurred making the pickle spoiled. - to prevent, we can use, turmeric powder, oil, salt, vinegar, etc.	1 1

खण्ड “घ” Section “D”		
17	- the negative impacts of digital technology; a) Effects in mental health b) cyber crime c) social isolation d) Physical effects on body - use of ICT for being good digital citizen a) polite and respective language in digital world b) respecting people’s privacy c) unnecessary and fake information should be avoided d) using authentic and trustworthy sites (1 mark for each correct effect maximum 2 marks)	1 1 1 1
18	a) it states that “ When cross is made between two pure individual having one pair of contrasting character, only one character is expressed in first filial generation, that is the dominant character. b)  phenotype ratio: 3:1 genotype ratio:1:2:1	1 2 1
19	a) A- RBC, C- Platelets b) Fibrinogen c) Person suffers from leukemia / Blood cancer	2 1 1
20	a) in moon there is no atmosphere, so there will be no air resistance to resist falling of parachute. b) Given; $M_1 = 2\times 10^{30}$Kg $M_2 = 1.9 \times 10^{27}$ Kg $d = 1.8 \times 10^{8}$Km $G = 6.67\times10^{-11}$ Nm^2/Kg^2 we have, $F = \frac{GM_1M_2}{d^2}$	1 2

	$F = \frac{6.67\times10^{-11}\times2\times10^{30}\times1.9\times10^{27}}{(1.8\times10^{11})^2}$ $F = 7.822\times10^{24}N$ Again; Now, Distance (d)= 3.6×10^{8}km. Solution; As the distance is made double than the previous, the force of attraction decreases four times. So, $F = 1.955\times10^{24}N$	1
21	a)  f = 4 cm = 0.04m p=? we have, p=1/f p= 1/0.04 p= 25D b) Concave lens 	1 1 1 1

<table>
<tr><td>22</td><td>a)
<table><tr><td>Alternating Current</td><td>Direct Current</td></tr><tr><td>a) Polarity changes
b) it is produced by dynamo and generator
c) it can be easily converted to DC
d) it can be transmitted to long distance.</td><td>a) Polarity does not change
b) it is produced by cell
c) it is difficult to convert to AC.
d) it is hard to transmit to long distance.</td></tr></table></td><td>2 mark for any two</td></tr>
<tr><td></td><td>b) Two ways to increasing the electricity from dynamo:
- by increasing the strength of magnet
- by increasing the number of turns in coil
- by decreasing the distance between magnet and coil
- by increasing the rotational speed of magnet</td><td>2 mark for any two</td></tr>
<tr><td>23</td><td>a) Carbondioxide (CO_2)
$CaCO_3$+ 2HCl----- $CaCl_2$ + H_2O + CO_2

b) H_2CO_3
c) by using moist blue litmus paper / by inserting burning matchstick / by reacting the gas with lime water.</td><td>1
1

1

1</td></tr>
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